

UNIT-I
Importance of Information System
CAP404
INFORMATION SECURITY AND PRIVACY

SCHOOL OF COMPUTER APPLICATIONS

DOMAIN: AI & ML

Importance of Information Systems

- **Improves decision-making** by providing accurate and timely information.
- **Increases efficiency through automation** of routine tasks and processes.
- **Enhances communication** within organizations and with customers or partners.
- **Supports business operations** such as inventory, sales, finance, and employee management.
- **Provides a competitive advantage** by improving productivity, innovation, and customer service.

1. Improves Decision-Making

CLASS EXPLANATION (Clothing Store)



Imagine you own a clothing store. Every day, the system records which shirts, jeans, and shoes are sold.

1. DATA COLLECTION

Sales are recorded every day.



2. WEEKLY REPORT

System shows sales summary.

ITEM	SOLD (WEEK)
Blue Shirts	500
Green Shirts	50
Jeans	300
Shoes	200

3. BETTER DECISION

Order more of what sells more.



Without an information system, you would have to guess.
With it, you know what customers want and can decide confidently.

REAL-WORLD EXAMPLE (Amazon)

amazon

Amazon tracks millions of customer purchases every day.



1. CUSTOMER PURCHASE DATA

Many people in a city start buying umbrellas.



2. INFORMATION SYSTEM ANALYSIS

The system identifies the trend and predicts higher demand for umbrellas in that city.



3. SMART DECISION

Amazon sends more umbrella stock to warehouses near that city.



4. BETTER RESULT

Customers get what they need on time. Sales and customer satisfaction increase.



Information Systems help Amazon (and other companies) make the right decisions at the right time – improving sales, reducing costs, and keeping customers happy.

2. Increases efficiency through automation

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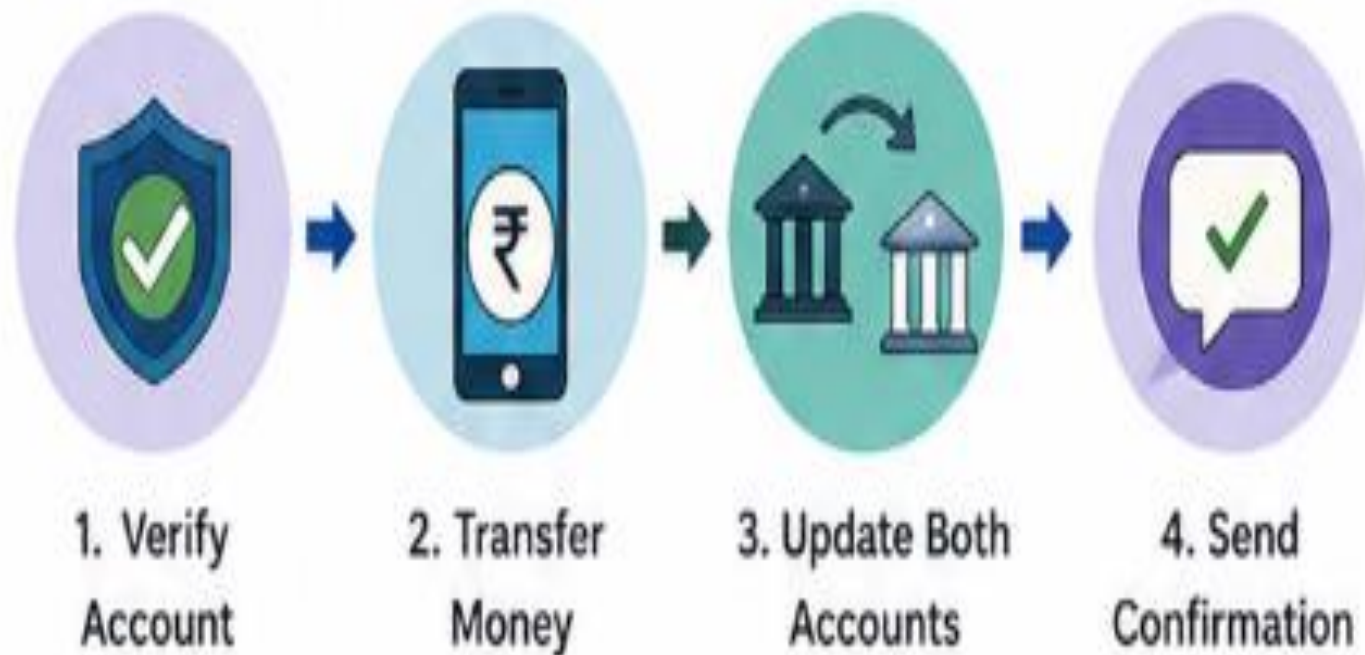
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REAL-WORLD EXAMPLE – PAYTM / UPI PAYMENT

When you pay ₹500 through Paytm or any UPI app, the system automatically does everything in just a few seconds.



All this happens in just a few seconds!



No bank employee manually processes each payment.

3. Enhances Communication

CLASS EXPLANATION – COLLEGE MANAGEMENT SYSTEM

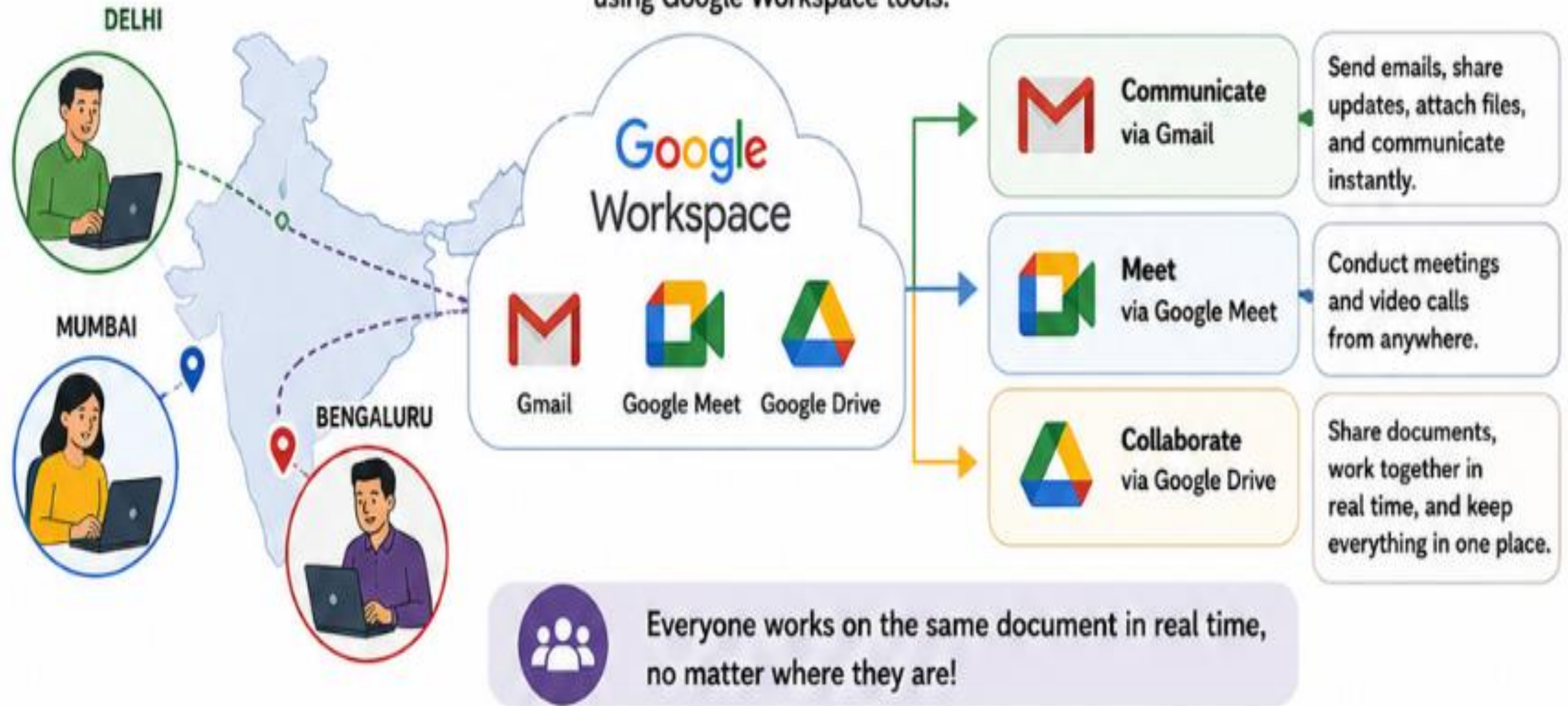
In a college, faculty members, students, and the administration need to communicate continuously.
A college management system can send important information instantly.



Everyone stays informed, on time, and connected!

REAL-WORLD EXAMPLE – GOOGLE WORKSPACE

A company with employees in Delhi, Mumbai, and Bengaluru can communicate and collaborate easily using Google Workspace tools.



4. Supports Daily Business Operations

CLASS EXPLANATION – SUPERMARKET EXAMPLE

Consider a supermarket. Every time a customer buys a product, the barcode is scanned. The Information System handles everything automatically.

1 Barcode Scanned



Product barcode is scanned at the billing counter.

2 Sale Recorded



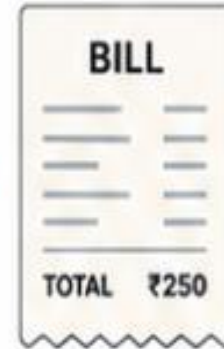
The system records the sale and updates the transaction.

3 Inventory Updated



Inventory is updated in real time.

4 Bill Generated



The system generates the bill for the customer.

5 Data Stored



Transaction is stored in the company database.

6 Low Stock Alert



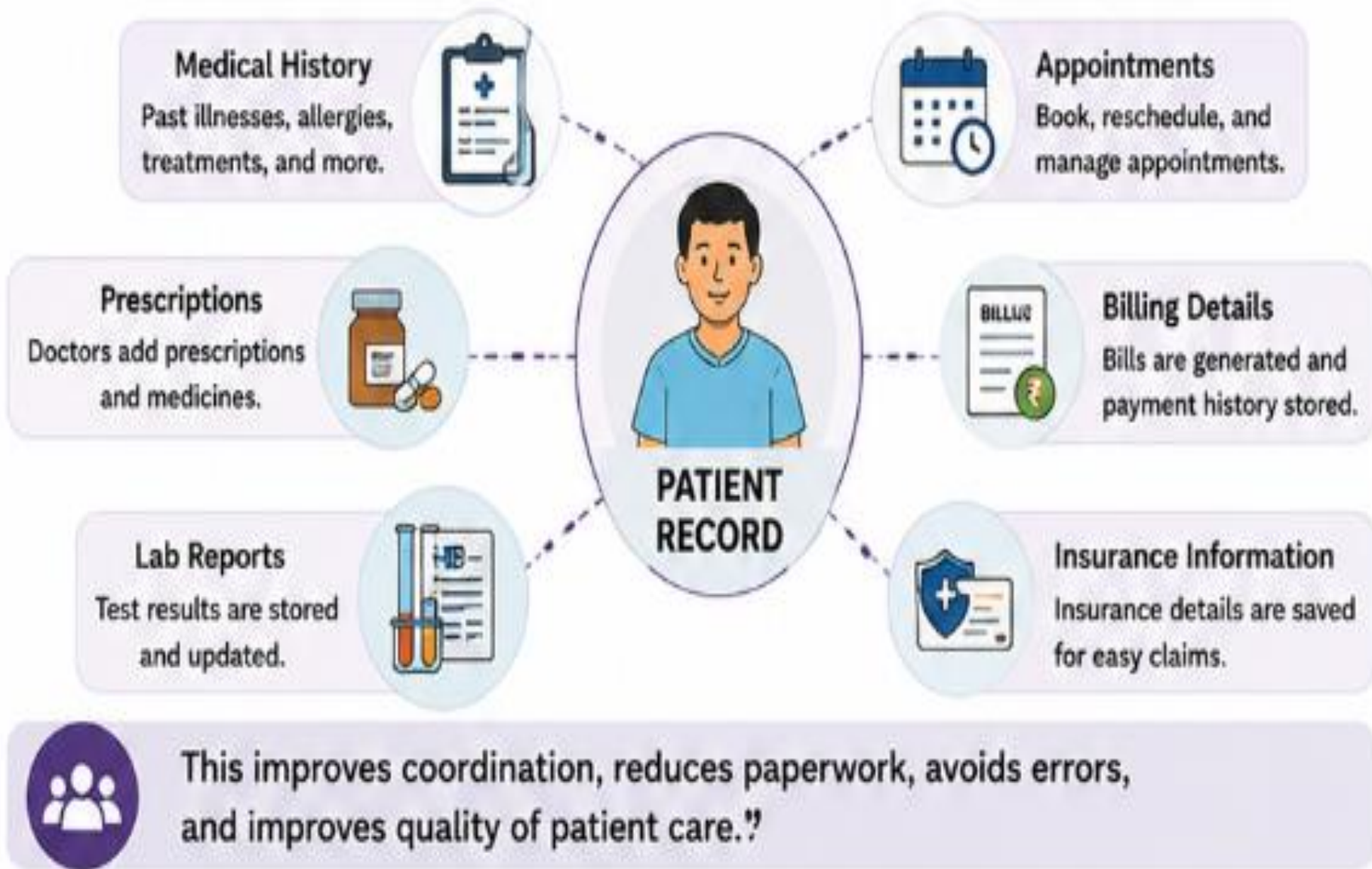
If stock is low, the system alerts the manager.



This saves time, reduces errors, prevents stock-outs, and helps management make better decisions.

REAL-WORLD EXAMPLE – HOSPITAL INFORMATION SYSTEM

When a patient visits a hospital, the Information System manages and connects all important data.



Doctors can access complete patient records instantly, anytime, anywhere.

5. Provides Competitive Advantage

CLASS EXPLANATION – FOOD DELIVERY EXAMPLE

Suppose two food delivery companies operate in the same city.

COMPANY A – WITHOUT ADVANCED IS



- ✗ No accurate delivery time prediction
- ✗ No suggestion of nearby restaurants
- ✗ No optimized route – longer delivery time
- ✗ Customers may get late delivery and get unhappy

VS

COMPANY B – WITH ADVANCED INFORMATION SYSTEM

- ✓ Predicts accurate delivery time 
- ✓ Shows nearby restaurants 
- ✓ Optimizes route in real-time 
- ✓ Faster delivery
- ✓ Happy customers 



Result: Company B delivers faster, has happier customers, and gets more orders.
This gives it a competitive advantage over Company A.

REAL-WORLD EXAMPLES

1 GOOGLE

Smart Search. Better Results.



information systems



Understands
what you mean



Uses AI & Information
Systems to find the most
relevant results



Delivers accurate results
in fractions of a second



Users get the right information quickly,
so Google stays the world's most
popular search engine.

2 AMAZON RECOMMENDATIONS

Personalized. Relevant. Engaging.



Customers who bought
this item also bought...



Collects & analyzes
customer behavior
(searches, views,
purchases)



Information System
finds patterns &
predicts what you
might like



Shows personalized
recommendations
to increase sales and
customer satisfaction



More engagement, more sales,
and loyal customers – a strong
competitive advantage for Amazon.

FEATURES OF INFORMATION SYSTEMS (IS)



1 DATA COLLECTION



Collects data from various sources

2 DATA PROCESSING



Processes raw data into meaningful information

3 DATA STORAGE



Stores data securely for future use

4 INFORMATION RETRIEVAL



Retrieves required information quickly when needed

5 ACCURACY & RELIABILITY



Ensures accurate, consistent and reliable information

6 AUTOMATION



Automates repetitive tasks and reduces manual effort

7 TIMELY INFORMATION



Provides up-to-date information at the right time



Which sequence correctly represents the logical flow of information within most Information Systems?

- A. Retrieval → Collection → Storage → Processing
- B. Collection → Processing → Storage → Retrieval
- C. Processing → Collection → Retrieval → Storage
- D. Storage → Collection → Processing → Retrieval

A university system automatically generates attendance reports from daily biometric scans. Which combination of features is involved?

- A. Data Collection and Data Processing
- B. Data Storage and Information Retrieval
- C. Automation and Timely Information only
- D. Accuracy & Reliability only

Which feature enables organizations to respond rapidly to changing market conditions?

- A. Data Storage
- B. Timely Information
- C. Information Retrieval
- D. Data Collection

Which statement best distinguishes Data Processing from Information Retrieval?

- A. Processing stores data permanently, while retrieval deletes data.
- B. Processing transforms raw data into information, while retrieval accesses stored information when needed.
- C. Processing collects data from users, while retrieval validates data accuracy.
- D. Processing automates tasks, while retrieval provides timely information only.